

Delta Lake MERGE Implementation

Week 7 Assignment Summary

Objective

Perform incremental data processing on a customer dataset using Delta Lake, and compare two common merge strategies: SCD Type 1 (overwrite) and SCD Type 2 (full history).

What was done

1. Loaded customer_master.csv into a Pandas DataFrame and inspected it (shape, dtypes, nulls).
2. Cleaned the data: removed 6 exact duplicate rows, filled missing city/segment values, dropped rows with a missing email.
3. Wrote the cleaned data into a Delta table using the deltalake (delta-rs) Python library.
4. Loaded customer_incremental.csv, a simulated batch with 15 updates to existing customers and 20 brand-new customers, and cleaned it the same way.
5. Applied an SCD Type 1 MERGE: matched customers were updated in place, unmatched customers were inserted as new rows.
6. Applied an SCD Type 2 MERGE: matched customers had their existing row closed out (is_current = False, end_date set) and a new versioned row inserted, preserving full history.
7. Validated the results by checking row counts against the expected distinct customer count and confirming there were no duplicate active customer_id values.
8. Displayed the final tables and a short summary of the results.

Results

Metric	Value
Master rows (raw)	106
Master rows (after cleaning)	95
Duplicate rows removed	6
Incremental rows	35 (15 updates + 20 new)
SCD1 final row count	117
SCD1 duplicate customer_id count	0
SCD2 total rows (incl. history)	130
SCD2 current rows	117
All validation checks	PASSED

Key takeaway

SCD Type 1 is simpler and good enough when only the current state of a customer matters. SCD Type 2 costs a bit more storage and logic but is worth it whenever historical accuracy or auditing is needed -- for example, to answer "what was this customer's segment on a given date", which SCD1 cannot answer once a value has been overwritten.